ADVENTIST UNIVERSITY OF CENTRAL AFRICA

Big Data Essentials — Midterm Project

MSc in Big Data Analytics

Student Name	Student ID
Rubagumya Brave	101392
Uwizeye Christelle	101465
Rwagasana Felicien	101364

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Lecturer: Dr. Kundan Kumar

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Big Data-Driven Road Safety Intelligence:

A Prototype Rwanda Road Safety Intelligence System (RRSIS)

Abstract

Rwanda recorded 9,995 road accidents in 2023, including 761 fatal accidents, representing a 78 per cent increase in total accidents since 2018. Despite this scale, the underlying incident-level data is not publicly accessible, limiting systematic pattern analysis. This report presents the Rwanda Road Safety Intelligence System (RRSIS), a prototype big data system developed using the Hadoop Distributed File System (HDFS) as the storage layer and Apache Spark (PySpark) 3.4.4 as the analytics engine. Using the Kaggle Road Accident Dataset as a surrogate dataset of 307,955 cleaned records across 23 columns, RRSIS identifies where, when and under what circumstances accident risk is highest. The system delivers a two-dimensional temporal analysis separating frequency from severity, an Accident Severity Index using ordinal weights (Slight=1, Serious=3, Fatal=5), a dangerous factor combination analysis across six simultaneous factors, a window-function location ranking by geographical category, and a composite Road Safety Risk Score combining frequency, severity and fatal count with min-max normalisation. Key findings include that night hours carry the highest severity ratio (1.511) despite the lowest accident frequency, that 74.9 per cent of all accidents occur on single carriageway roads, and that the most dangerous factor combinations occur in fine weather rather than adverse conditions. Five ranked management priorities are produced with full numerical evidence, directly applicable to Rwanda road safety policy once access to Rwanda National Police incident-level data is granted.

Contents

1 Introduction and Literature Review

1.1 Background — Rwanda Road Safety Crisis

The NISR Statistical Yearbook 2024, Table 14.2.6, records 9,995 road accidents in Rwanda in 2023, of which 761 were fatal, and Rwanda National Police [6] records 723 fatal victims in that year.

Table 1: Road accidents recorded in Rwanda by severity classification, 2018–2023

Year	Fatal	Serious	Minor	Property	Total
2018	597	885	1,887	2,242	5,611
2019	673	911	1,485	1,584	4,653
2020	675	710	1,326	1,492	4,203
2021	621	471	3,688	3,859	8,639
2022	676	110	5,186	4,362	10,334
2023	761	262	5,037	3,935	9,995

Source: Rwanda National Police / NISR Statistical Yearbook 2024, Table 14.2.6

Total accidents rose from 5,611 in 2018 to 9,995 in 2023, a **78 per cent increase**, while fatal accidents never fell below 597 and peaked in the most recent year.

Rwanda collects accident data but has no system capable of interrogating it at scale, so no operational analysis identifies when accidents occur, where they concentrate, or under what circumstances they become severe. RRSIS bridges this gap using HDFS for distributed storage and Apache Spark (PySpark) for analytics, converting accident records into ranked management priorities supported by numerical evidence.

1.2 Problem Statement

Rwanda records thousands of road accidents annually through Rwanda National Police, and aggregate statistics are published through NISR. However, the underlying incident-level data is not publicly accessible at the scale and attribute richness required for systematic pattern analysis. No operational data intelligence infrastructure exists to interrogate these records in a way that identifies where, when and under what circumstances accident risk is highest. This gap creates three problems: data volume, since manual cross-tabulation of thousands of records across multiple attributes is not feasible; pattern identification, since high-risk times, locations and dangerous conditions remain unidentified; and evidence-based decisions, since interventions lack numerical justification for resource allocation.

To bridge this gap, RRSIS uses the Kaggle Road Accident Dataset as a surrogate dataset — a large, attribute-rich, openly licensed dataset that permits the full RRSIS pipeline to be built, demonstrated and validated. The patterns derived represent universal road safety dynamics directly transferable to the Rwanda context. RRSIS is therefore framed as a descriptive prioritization tool and a methodological framework ready to be applied to actual Rwanda National Police incident-level data once access is granted.

1.3 Study Objectives

- **Objective 1:** Implement a distributed storage and processing pipeline using HDFS and Apache Spark.
- **Objective 2:** Identify temporal patterns in accident frequency and severity.
- **Objective 3:** Develop an Accident Severity Index ranking locations, road types and vehicle types.
- **Objective 4:** Identify dangerous factor combinations associated with highest severity.
- **Objective 5:** Produce a composite Road Safety Risk Score ranking locations for intervention.

1.4 Significance of the Study

Existing literature relied on 1,536 to 30,358 records on single workstations, whereas RRSIS processes 307,955 records on distributed infrastructure. The five management priorities deliver ranked, evidence-based intervention areas that Delen et al. (2006) variable importance tables and Mujalli and de Oña (2011) conditional probabilities do not provide. With 9,995 accidents in 2023 and a 78 per cent five-year increase, Rwanda needs exactly this intelligence to reverse the trend.

1.5 Purpose and Use of the Surrogate Dataset

The analysis uses the Kaggle Road Accident Dataset [8] as a surrogate dataset. United Kingdom road accident records from 2021–2022 comprising 307,973 records across 23 columns. It was selected because it is large enough to justify a distributed processing architecture, attribute-rich enough for multi-factor analysis, and openly licensed for publication and reproduction. Although the records are not Rwanda-specific, the mechanisms that make a road dangerous, reduced visibility, driver fatigue, speed differentials, absence of traffic separation are not nationally specific, and the patterns the dataset reveals represent universal road safety dynamics applicable to the Rwanda context.

Four areas of insight transfer directly: night-time severity, single carriageway road risk, evening peak concentration, and the role of driver behavior over weather conditions. These are discussed in detail in Section 4.9. The RRSIS framework applies directly to Rwanda National Police and NISR incident-level data without modification to its architecture, pipeline or scoring methodology, once access to that data is granted.

1.6 Literature Review

1.6.1 Abdelwahab and Abdel-Aty (2001)

Abdelwahab and Abdel-Aty (2001) [1] used neural networks on 1,536 records and found time of day ($p = 0.532$), day of week ($p = 0.250$) and weather ($p = 0.556$) insignificant for severity, since "drivers may take appropriate measures in bad weather conditions", while their Mussone accident index anticipates the normalised frequency component of the RRSIS risk score.

1.6.2 Mujalli and de Oña (2011)

Mujalli and de Oña (2011) [3] used Bayesian networks with selection across 59 algorithm combinations, found time of day selected only 15 times and day of week 18 with neither in the final BN-7 model, showed P(KSI) escalating from 58.91 to 96.51 per cent across seven combined factors, and reported good weather more severe (0.4653) than heavy rain (0.3124).

1.6.3 Delen, Sharda and Bessonov (2006)

Delen et al. (2006) [2] used neural networks with sensitivity analysis on 30,358 records in which fatalities were 3.6 per cent, supplied the RRSIS min-max normalisation as Equation 1, and stated that "no single factor by itself appears to be a key determinant of accident severity, but can act as a catalyst or a barrier in combination with other factors".

1.6.4 Savolainen, Mannering, Lord and Quddus (2011)

Savolainen et al. (2011) [7] established that "injury-severity levels are ordinal by nature", that fatal reporting is "nearly 100 percent" complete while slight injuries are undercounted 50–75 per cent, that spatially proximate crashes "share unobserved effects", and that frequency and severity reduction "may necessitate different strategic approaches".

1.6.5 Rahm and Do (2000)

Rahm and Do (2000) [5] define data cleaning as "detecting and removing errors and inconsistencies from data", warn of "garbage in, garbage out", and supply the five-phase process and 2×2 taxonomy used to classify all eight quality issues in Section 3.2.

1.7 Three Cross-Paper Findings

Temporal variables are weak severity predictors, established by Abdelwahab and Abdel-Aty (2001), Mujalli and de Oña (2011) and Delen et al. (2006), so RRSIS frames temporal results as exposure rather than severity explanation. **Adverse weather does not increase severity**, being insignificant at $p = 0.556$, inverted at P(KSI) 0.4653 against 0.3124, and excluded from all models respectively. **Severity is driven by combinations, not single factors**, stated by Delen et al. (2006), quantified by Mujalli and de Oña (2011) and simulated across 9,216 patterns by Abdelwahab and Abdel-Aty (2001).

1.8 Positioning RRSIS

RRSIS is a **descriptive prioritization tool**, not a predictive model. This means it does not forecast future accidents or estimate probabilities, it analyses what has already happened and ranks where intervention is most needed. Every number in this report comes directly from counting and weighting observed accident records, making the results transparent and auditable. Compared to existing literature, RRSIS makes three contributions: it processes 307,955 records against a maximum of 30,358 in any reviewed paper; it retains 99.994 per cent of records through careful data cleaning, against losses of 9.5 to 53 per cent in the literature; and it delivers five ranked management interventions with numerical evidence, rather than statistical tables that require further interpretation before action can be taken.

2 System Architecture and Dataset

2.1 System Architecture

RRSIS is built on two layers working together. The first layer is HDFS, which stores the dataset at `/Group_15/midterm_dataset/` and ensures the data is distributed, replicated and protected against loss. The second layer is Apache Spark 3.4.4, which reads from HDFS and processes the data in parallel across multiple partitions, making it fast enough to handle 307,955 records efficiently. The data flows through a simple pipeline: Raw Data → HDFS → Spark → Cleaning → Analytics → Risk Model → Management Decisions. Each step feeds the next, and the cleaned dataset is reused across all nine analytical tasks without being read from HDFS again, saving processing time. Following the principle of Rahm and Do (2000) [5], the original raw CSV file is never modified. This means every result in this report can be traced back to the unmodified source, and the entire analysis can be re-run at any time to verify or update the findings.

2.2 Dataset Description

The surrogate dataset is a single CSV file of 72.2 MB containing 307,973 road accident records across 23 columns, covering United Kingdom accidents in 2021–2022. Each record represents one accident rather than one casualty, meaning that a single accident involving multiple injured people appears as one row — consistent with the severity definition used by Mujalli and de Oña (2011) [3]. The dataset has a significant imbalance in severity: 85.49 per cent of accidents are slight, 13.23 per cent serious and only 1.28 per cent fatal. This means fatal accidents are rare in the data, which is an even more extreme imbalance than the 3.6 per cent reported by Delen et al. (2006) [2]. This imbalance is the main reason why a weighted severity index is needed — without it, the analysis would be dominated entirely by slight accidents and would undercount the most serious outcomes. It is important to note that the UK road network differs from Rwanda's. The UK has more dual carriageway roads and a fleet dominated by cars, while Rwanda has proportionally more single carriageway roads and a much larger share of motorcycles. The numbers reported in Section 4 therefore demonstrate how the RRSIS method works, rather than providing direct estimates of Rwandan road safety quantities.

Table 2: Key columns used in the RRSIS analysis

Column	**Role**	**Analytical use**
Accident_Severity	Outcome	Severity index weighting (Tasks 4–7)
Accident Date	Temporal	Day of week, month extraction (Task 3)
Time	Temporal	Hour extraction, time period classification (Task 3)
Local_Authority_(District)	Geographic	Location ranking, risk score (Tasks 4, 6, 7)
Urban_or_Rural_Area	Geographic	Window partition key (Task 6)
Road_Type	Infrastructure	Severity by road type, combinations (Tasks 4, 5)
Speed_limit	Infrastructure	Factor combination analysis (Task 5)
Road_Surface_Conditions	Environmental	Factor combination analysis (Task 5)
Weather_Conditions	Environmental	Factor combination analysis (Task 5)
Carriageway_Hazards	Environmental	Quality treatment (Task 2)
Vehicle_Type	Vehicle	Severity by vehicle type (Task 4)

Table 3: Accident severity distribution in the cleaned surrogate dataset

Severity class	**Accidents**	**Share of total**
Slight	263,264	85.49%
Serious	40,738	13.23%
Fatal	3,953	1.28%
Total	**307,955**	**100.00%**

3 Methodology

3.1 Data Ingestion

Data ingestion was completed in three steps. First, a dedicated project directory was created on HDFS using the command `hdfs dfs -mkdir -p /Group_15/midterm_dataset/`. Second, the CSV file was uploaded from the local machine using `hdfs dfs -put` and confirmed at 72.2 MB. Third, the dataset was loaded into a PySpark DataFrame by reading directly from the HDFS path, using `header=True` to treat the first row as column names and `inferSchema=True` to automatically detect column data types. The `inferSchema=True` setting caused one unintended issue: the Time column in the raw data contained only clock times in HH:mm format (for example, 17:30). Spark did not recognise these as simple time values and instead converted them into full timestamps by adding the current session date, producing values like `2026-09-02 17:30:00` across every record. The date part was completely fictitious and identical for all rows. This was identified as a data quality issue and corrected in the cleaning stage described in Section 3.2. Evidence screenshots of the HDFS directory creation, file upload, schema and record count are available in the project GitHub repository at https://github.com/Rubagumyabrave/Midterm_project_group_15 (see Appendix A).

3.2 Data Quality Engineering

The assignment requires identifying missing values, duplicate records, inappropriate data types and at least five data quality issues, cleaning them using Spark DataFrame operations and justifying each treatment. RRSIS satisfies all these requirements and structures the cleaning process within the five-phase framework of Rahm and Do (2000) [5]: (1) analysing the data for problems, (2) defining how each problem will be fixed, (3) verifying the fixes work, (4) applying the fixes, and (5) writing the cleaned data back to the original source.

RRSIS fully implements phases 1, 2 and 4. Phase 3 was carried out by re-running the same quality checks after cleaning to confirm all issues were resolved. Phase 5 was not applicable because the source is a static third-party CSV file that cannot be modified.

Eight data quality issues were identified — exceeding the minimum of five required — and each was classified using the Rahm and Do 2×2 taxonomy, which categorises problems by origin (single source or multiple sources) and nature (schema-level affecting structure, or instance-level affecting individual records). Table 4 summarises all eight issues, their classification, the treatment applied and the number of records affected.

Table 4: Data quality issues identified and treated, classified against the Rahm and Do (2000) taxonomy

#	Issue	Rahm & Do classification	Treatment	Records affected
1	Accident Date stored as String rather than DateType	Single-source, schema-level	Converted using to_date()	All 307,973
2	Time carries fabricated date 2026-09-02 introduced by inferSchema=True	Single-source, schema-level	Extracted HH:mm using date_format()	All 307,973
3	Weather_Conditions missing values	Single-source, instance-level	Filled with “Unknown”	6,057
4	Road_Type missing values	Single-source, instance-level	Filled with “Unknown”	1,534
5	Road_Surface_Conditions missing values	Single-source, instance-level	Filled with “Unknown”	317
6	Carriageway_Hazards missing values	Single-source, instance-level	Filled with “None”	3
7	Time missing entirely	Single-source, instance-level	Rows dropped	17
8	Duplicate record	Single-source, instance-level	Removed using dropDuplicates()	1

For columns with missing values, the decision was made to fill them in rather than delete the entire row. Deleting rows is the easiest option but it introduces bias, the 6,057 rows with missing weather information represent nearly two per cent of the dataset, and those rows are not missing randomly. Removing them would silently distort every analysis that touches weather, location or time. Instead, missing values in Weather_Conditions, Road_Type and Road_Surface_Conditions were filled with ‘Unknown’, making the missingness visible in any analysis that conditions on those fields rather than hiding it through deletion.

Carriageway_Hazards was treated differently, filled with ‘None’ rather than ‘Unknown’, because this field records hazards by exception. If no hazard was recorded, the most plausible interpretation is that no hazard was present, not that the officer forgot to check.

The only column where deletion was justified was Time. A missing accident time cannot be guessed or estimated without fabricating data, and time is the foundation of the entire temporal analysis in Section 4.3. Only 17 rows were dropped for this reason. The final cleaned dataset contains 307,955 records from an original 307,973, a loss of just 0.006 per cent and a retention rate of 99.994 per cent. This compares favourably with the reviewed literature, where cleaning losses range from 9.5 to 53 per cent due to listwise deletion of any row with a missing value.

3.3 Temporal Analysis

Every temporal unit is characterised by both accident frequency and severity ratio, following Savolainen et al. (2011) [7], who observe that "reducing crash frequency and reducing crash-injury severity may necessitate different strategic approaches". The severity ratio is Severity_Score / Total_Accidents, bounded on [1, 5] and interpretable as the mean severity weight of an accident in that unit, and because it is normalised by exposure it is comparable across periods of very different traffic volume.

Table 5: Time period classification scheme

Period	**Hour range**	**Rationale**
Night	00:00–05:59	Darkness, minimal traffic, fatigue and impairment exposure
Morning	06:00–11:59	Morning commute and commercial activity
Afternoon	12:00–15:59	Midday movement, school transport
Evening	16:00–19:59	Evening commute peak
Late Night	20:00–23:59	Post-evening social travel, declining traffic volume

3.4 Accident Severity Index

The index applies weights **Slight = 1, Serious = 3, Fatal = 5**, giving Severity Score = Σ(count $\times$ weight), justified by three literature sources: ordinality (Savolainen et al., 2011), class imbalance at 3.6 per cent fatalities (Delen et al., 2006), and reporting bias with fatal reporting near 100 per cent against slight injuries undercounted 50–75 per cent (Savolainen et al., 2011). The weights are assigned by judgement rather than estimated from data, a limitation acknowledged in Section 5.2, though the scheme is transparent, arithmetically traceable and applied uniformly across every grouping.

3.5 Dangerous Factor Combination Analysis

Road accidents are rarely caused by a single factor — it is the combination of conditions that determines how serious an outcome will be. Delen et al. (2006) [2] state this explicitly: no single factor is by itself a key determinant of accident severity, but each one "can act as a catalyst or a barrier in combination with other factors." Task 5 puts this principle into practice.

Six factors are analysed together in a single Spark `groupBy` operation: Road_Type, Speed_limit, Weather_Conditions, Road_Surface_Conditions, Time_Period and Vehicle_Type. This means every unique combination of these six factors is treated as one group, and the total severity score for each combination is calculated. For example, 'Single carriageway + 30mph + Fine weather + Dry surface + Evening + Car' is one combination, and its severity score tells us exactly how dangerous that specific set of conditions is.

Combinations are ranked by severity score rather than by the number of accidents. This is an important distinction, a combination that produces many minor accidents should not rank above one that produces fewer but more fatal or serious accidents. The severity score, using the weights Slight=1, Serious=3 and Fatal=5, ensures the ranking reflects actual harm rather than just frequency.

3.6 Window Function Location Ranking

Task 6 ranks the top three highest-risk locations separately for urban and rural areas. Ranking them separately is important, if all locations were ranked together, large urban authorities like Birmingham would dominate the entire list, hiding rural locations that may have equally dangerous road environments despite lower accident volumes.

The ranking uses a Spark Window function, which works like a filter that says: 'within each geographical category, order locations by severity score from highest to lowest and assign a rank.' Three ranking methods are available in Spark, and the choice matters:

- **row_number()** assigns a unique rank to every location, but if two locations have the same severity score it breaks the tie randomly, meaning the result could be different each time the code runs.
- **rank()** handles ties correctly but skips numbers after a tie, so if two locations share rank 2, the next location gets rank 4, and a filter for rank ≤ 3 would return only two locations instead of three.
- **dense_rank()** was chosen because it handles ties correctly and never skips ranks, a filter for rank ≤ 3 always returns exactly the top three locations, consistently and reproducibly.

One limitation applies: Savolainen et al. (2011) [7] warn that locations close to each other geographically often share the same underlying road conditions and enforcement environment. This means the top three ranked locations in one area may reflect one shared problem rather than three separate ones, and should be investigated together rather than treated as independent risks.

3.7 Road Safety Risk Score

The Risk Score combines three measures into a single number that captures how dangerous a location is from multiple angles: how often accidents happen there (frequency), how serious those accidents are (severity), and how many people have been killed there (fatal count).

Before combining them, each measure is converted to the same scale using min-max normalisation, a formula from Delen et al. (2006) [2]: $v' = (v - \text{minimum})/(\text{maximum} - \text{minimum})$. This transforms every value to a number between 0 and 1, where 0 represents the safest location and 1 represents the most dangerous. Without this step, combining the three measures would be meaningless, a raw accident count of 6,165 and a severity score of 7,805 cannot be added together directly because they are on completely different scales.

The three normalised measures are then combined using weighted addition:

$$\text{Risk Score} = (0.3\times\text{Normalised Frequency})+(0.4\times\text{Normalised Severity})+(0.3\times\text{Normalised Fatal Count})$$

The weights reflect the importance of each measure:

- **Severity (0.4)** carries the highest weight because it measures actual harm, not just how many accidents occurred, but how serious they were.
- **Fatal Count (0.3)** is given its own component because fatal accidents are extremely rare in the data (only 1.28 per cent of records) and would be overwhelmed by slight accidents without separate treatment, as Delen et al. (2006) demonstrate.
- **Frequency (0.3)** is included because a location that consistently generates many accidents still represents a systemic risk, following the location index principle of Abdelwahab and Abdel-Aty (2001).

The final Risk Score for every location falls between 0 and 1, making locations directly comparable regardless of their size or the volume of traffic they carry.

3.8 Spark Performance Analysis

To understand how Spark executes the analysis, `df.explain(True)` was run on the `severity_by_location` operation. This prints four plans that show what happens between writing the code and getting the result:

- **Parsed Plan**: what the code says, before Spark checks it
- **Analyzed Plan**: after Spark validates all column names and data types
- **Optimized Plan**: after Spark's Catalyst optimizer rearranges the steps to run as efficiently as possible
- **Physical Plan:** the exact execution steps Spark will actually perform

This demonstrates the principle Rahm and Do (2000) [5] describe: the analyst writes what they want (the intent), and Spark figures out the best way to do it (the execution). The analyst does not need to manage how the data is processed, Spark handles that automatically.

Spark also separates operations into two types. Transformations such as `filter()`, `groupBy()` and `withColumn()` are lazy, they do not run immediately but build up a list of instructions. Actions such as `count()` and `show()` trigger the actual execution. This means Spark can optimise the entire instruction list before running anything.

Because `df_cleaned` is used across all tasks from Task 3 to Task 9, `cache()` was applied after cleaning. Without `cache()`, every task would re-read the raw CSV from HDFS and re-run all eight cleaning steps from scratch each time. With `cache()`, the cleaned dataset is stored in memory after the first use, and all subsequent tasks read directly from memory, significantly reducing processing time.

4 Results and Analysis

4.1 Data Ingestion Results

After loading the dataset, Spark confirmed 23 columns and 307,973 records. Screenshots of the schema and record count are available on the project GitHub repository (https://github.com/Rubagumyabrave/Midterm_project_group_15) as `Task1_06_Dataset_Schema.JPG` and `Task1_07_Record_Colu`
Inspecting the schema revealed two problems introduced by automatic type detection. The Accident Date column was stored as text instead of a proper date type, and the Time column contained a fictitious date added by Spark during loading. Both issues are described and fixed in Section 3.2. This confirms an important lesson: when Spark automatically detects data types, the result must always be inspected before the data is used — automatic detection is a convenience, not a guarantee of correctness.

4.2 Data Quality Results

Table 6: Data quality outcomes, before and after cleaning

Measure	**Before cleaning**	**After cleaning**
Total records	307,973	307,955
Records with missing Weather_Conditions	6,057	0
Records with missing Road_Type	1,534	0
Records with missing Road_Surface_Conditions	317	0
Records with missing Carriageway_Hazards	3	0
Records with missing Time	17	0
Duplicate records	1	0
Accident Date type	String	DateType
Time representation	Timestamp with fabricated date	HH:mm string

After cleaning, a total of **7,928** missing values were filled in and one duplicate record was removed, leaving **307,955** records from an original 307,973, a retention rate of 99.994 per cent. Evidence screenshots of the missing value audit and the final record count are available on the project GitHub repository (https://github.com/Rubagumyabrave/Midterm_project_group_15) as `Task2_01_Missing_Values.JPG` and `Task2_05_After_Cleaning_Count.JPG`.

4.3 Temporal Analysis Results

4.3.1 Frequency Analysis Results

Table 7: Accidents by hour — five highest and lowest hour

Rank	Hour	Accidents
1	17:00	26,964
2	16:00	24,902
3	15:00	24,151
4	08:00	22,552
5	18:00	21,063
—	04:00 (lowest)	1,721

The accident distribution across the day follows a two-peak pattern. The main peak runs from 15:00 to 18:00, coinciding with the evening commute when traffic volume is at its highest. A secondary peak appears at 08:00, reflecting the morning rush hour. The quietest hour is 04:00 with only 1,721 accidents, compared to 26,964 at the busiest hour of 17:00, meaning accidents at the peak hour are approximately 16 times more frequent than at the quietest hour.

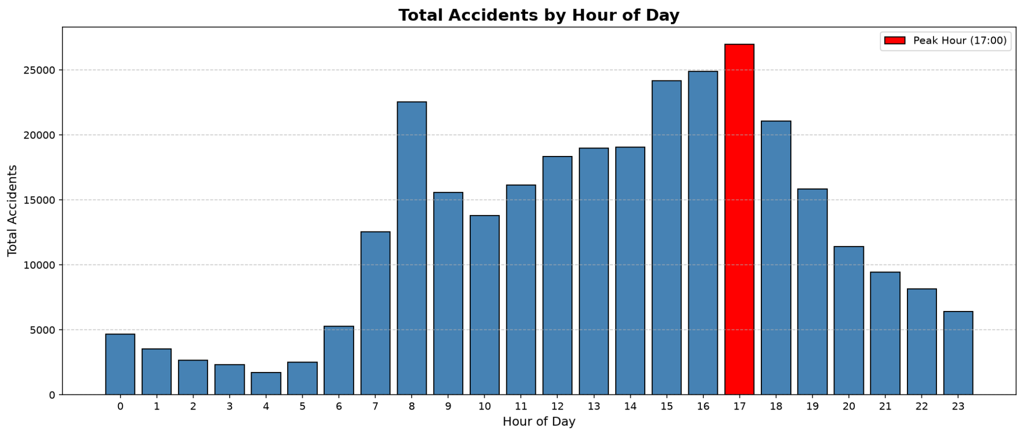

Figure 1: Total Accidents by Hour of Day — Peak at 17:00 with 26,964 accidents

Table 8: Accidents by day of week

Day	Accidents
Friday	50,529
Tuesday	46,382
Wednesday	46,378
Thursday	45,646
Monday	43,913
Saturday	41,564
Sunday	33,543
Total	**307,955**

Friday stands out clearly from the other weekdays. It records 50,529 accidents, 8.9 per cent more than Tuesday, the next highest day. By comparison, the difference between the remaining weekdays Monday to Thursday is much smaller, suggesting that Friday carries a distinct risk above the ordinary weekday pattern, likely driven by end-of-week fatigue and behavioral factors.

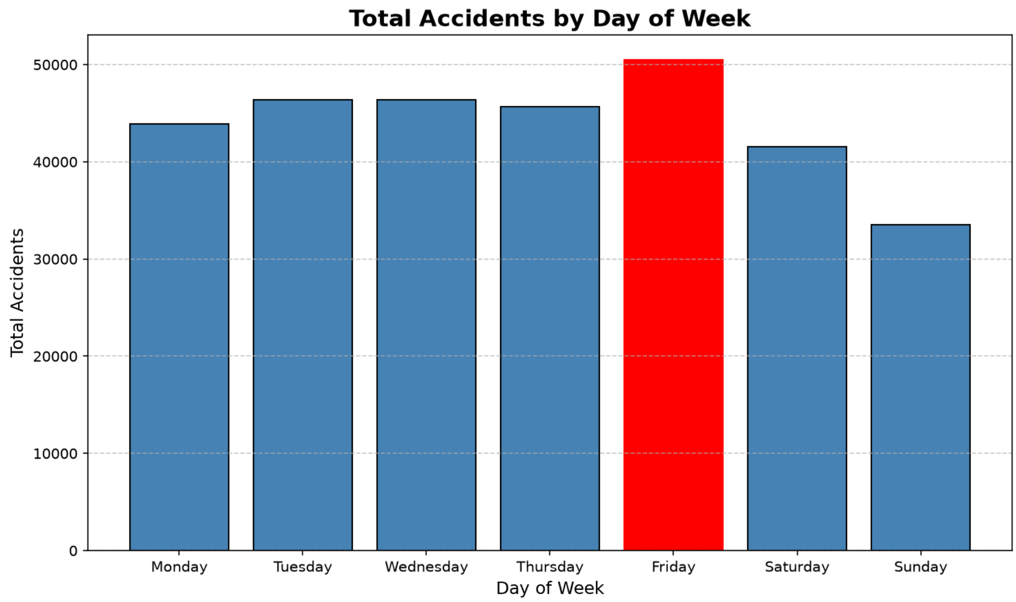

Figure 2: Total Accidents by Day of Week — Friday highest with 50,529 accidents

Table 9: Accidents by month — highest and lowest

Month	Accidents
November	29,094
October	28,368
July	26,952
February (lowest)	21,882

The higher accident counts in October and November are likely explained by shorter daylight hours. As the days get darker earlier in autumn, the evening commute, which peaks between 15:00 and 18:00, increasingly takes place after sunset, combining high traffic volume with reduced visibility and creating more dangerous driving conditions.

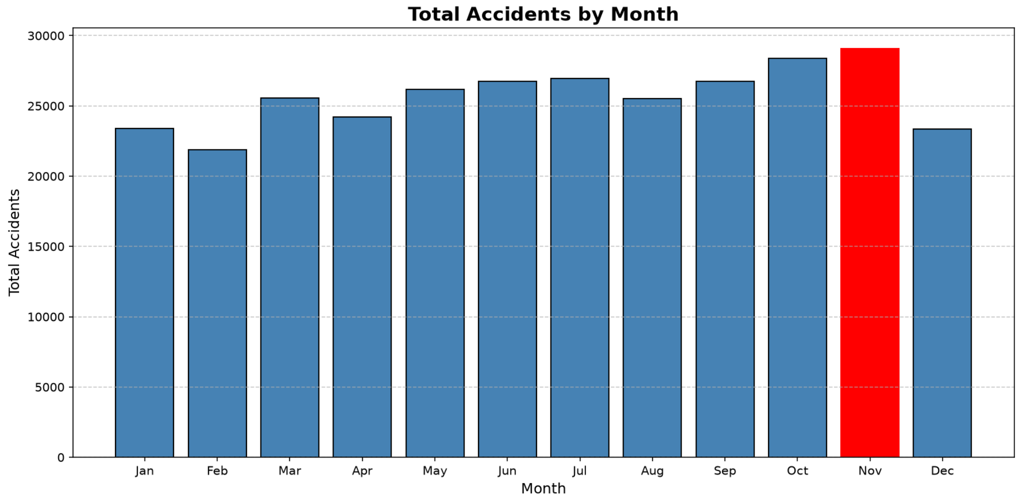

Figure 3: Total Accidents by Month — November highest with 29,094 accidents

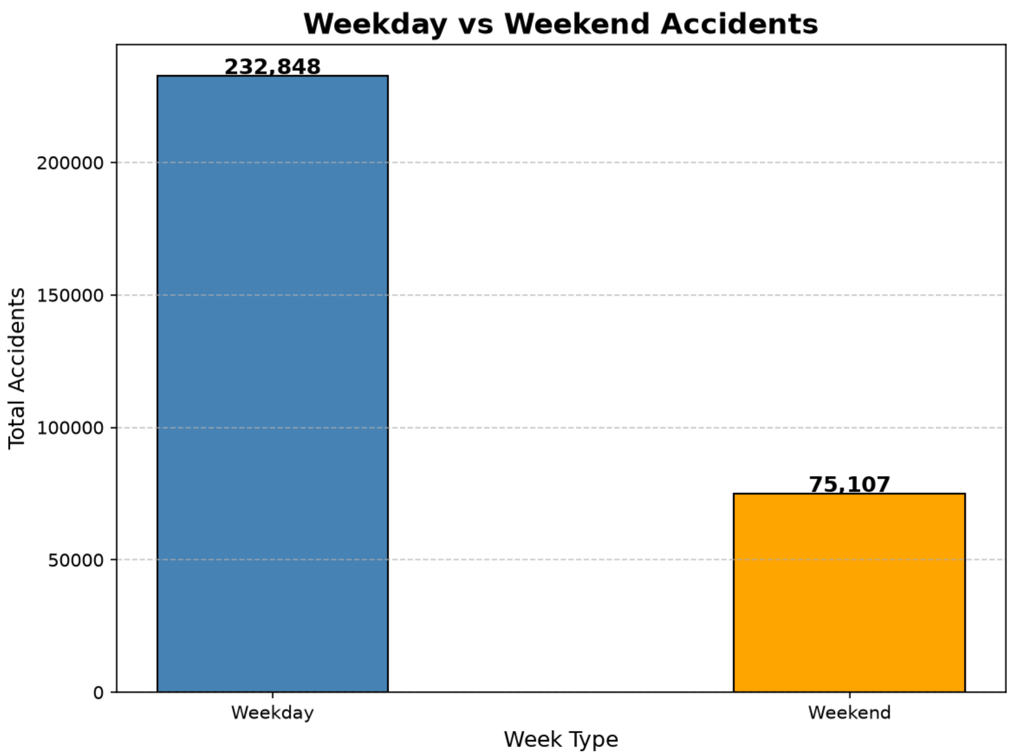

Figure 4: Weekday vs Weekend Accident Frequency — Weekdays 232,848 (75.6%) vs Weekends 75,107 (24.4%)

Table 10: Accidents by time period

Time period	**Hour range**	**Accidents**	**Share**
Evening	16:00–19:59	88,780	28.83%
Morning	06:00–11:59	85,878	27.89%
Afternoon	12:00–15:59	80,531	26.15%
Late Night	20:00–23:59	35,391	11.49%
Night	00:00–05:59	17,375	5.64%
Total		**307,955**	**100.00%**

If road safety resources were allocated purely based on accident frequency, the Night period would receive the smallest share, only 5.64 per cent of accidents occur between 00:00 and 05:59. However, as Section 4.3.2 shows, this would be a dangerous mistake, because night accidents are far more likely to be fatal than accidents at any other time of day.

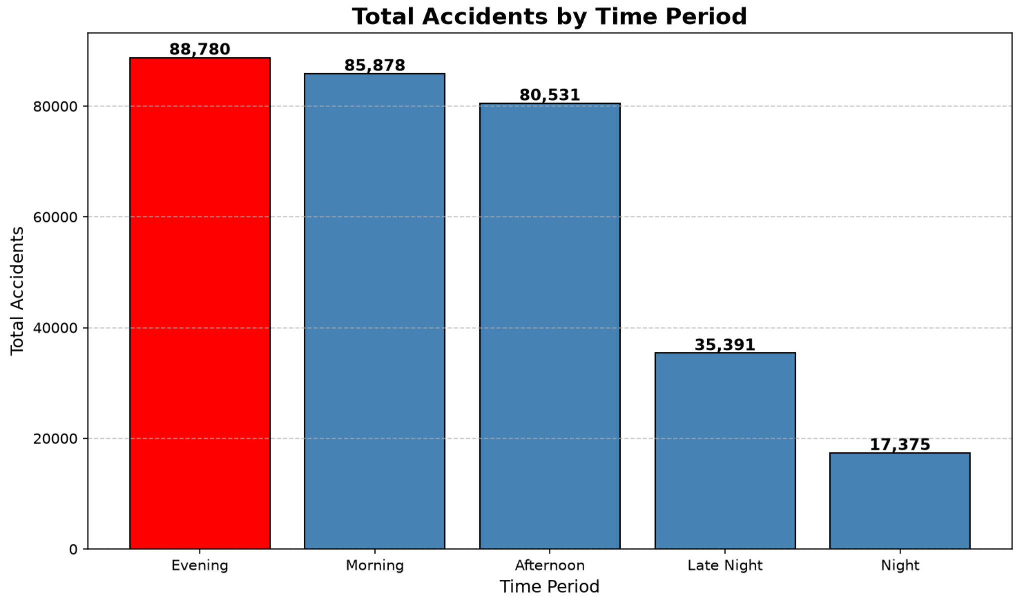

Figure 5: Total Accidents by Time Period — Evening highest with 88,780 accidents

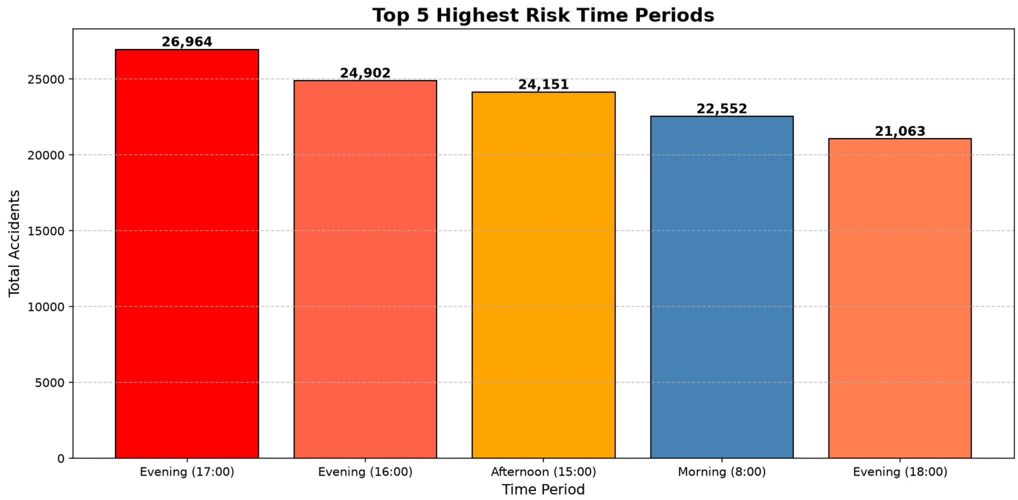

Figure 6: Top 5 Highest Risk Time Periods by Frequency

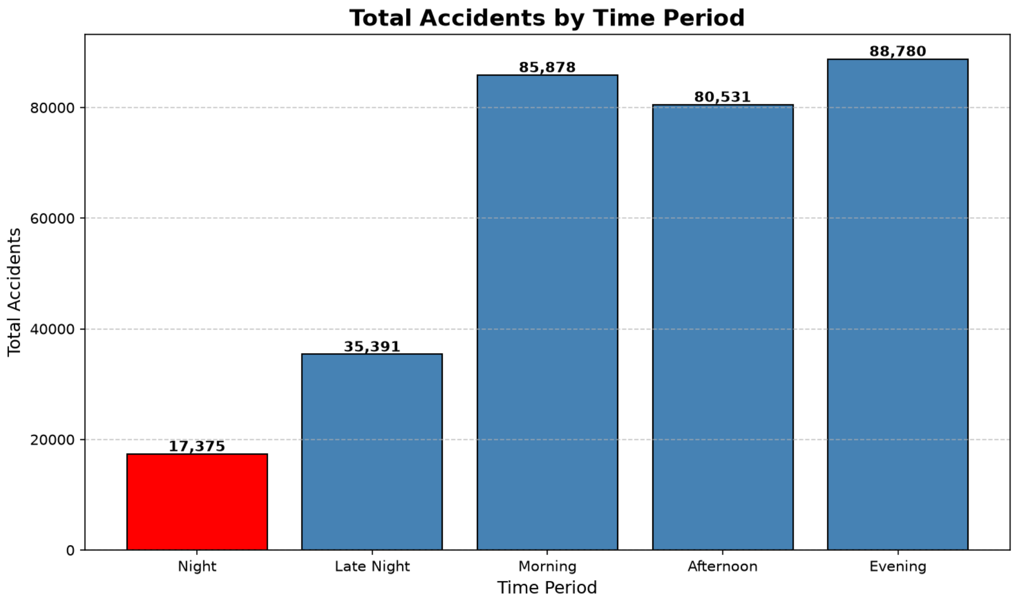

Figure 7: Total Accidents by Time Period

4.3.2 Severity Ratio Analysis Results

Table 11: Severity ratio, fatal rate and fatality count by time period

Time period	**Accidents**	**Severity ratio**	**Fatal rate**	**Fatalities**
Night	17,375	1.511	3.43%	596
Late Night	35,391	1.383	1.82%	643
Evening	88,780	1.307	1.06%	942
Afternoon	80,531	1.291	0.98%	792
Morning	85,878	1.282	1.14%	980
Total	**307,955**			**3,953**

The severity analysis reveals a striking reversal. The Night period, which has the fewest accidents of any period, is actually the most dangerous per accident, it records the highest severity ratio of 1.511 and a fatal rate of 3.43 per cent. This is more than three times the Evening fatal rate of 1.06 per cent, despite Evening having five times more accidents. In other words, the period that looks safest by frequency is the most deadly by severity, the two measures produce almost exactly opposite rankings.

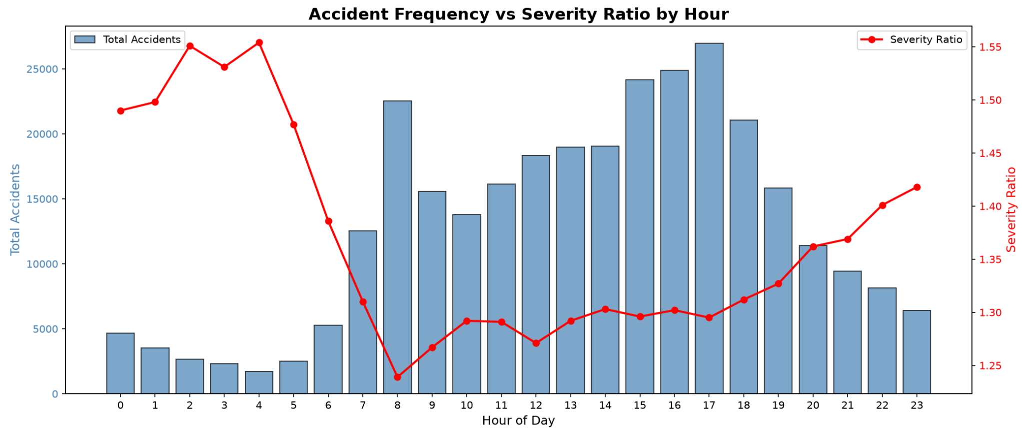

Figure 8: Accident Frequency vs Severity Ratio by Hour — Night hours show highest severity despite lowest frequency

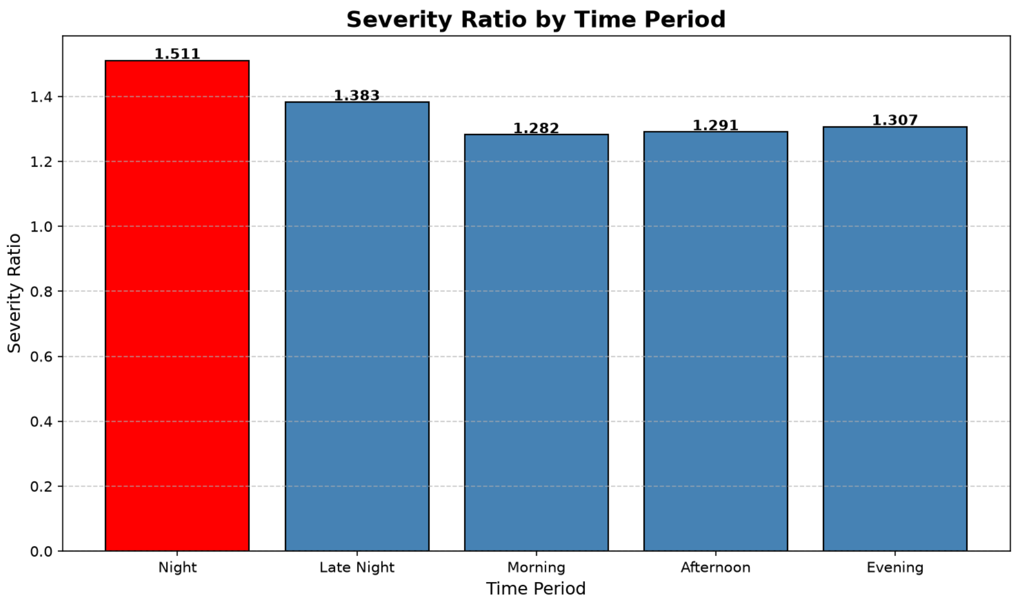

Figure 9: Severity Ratio by Time Period

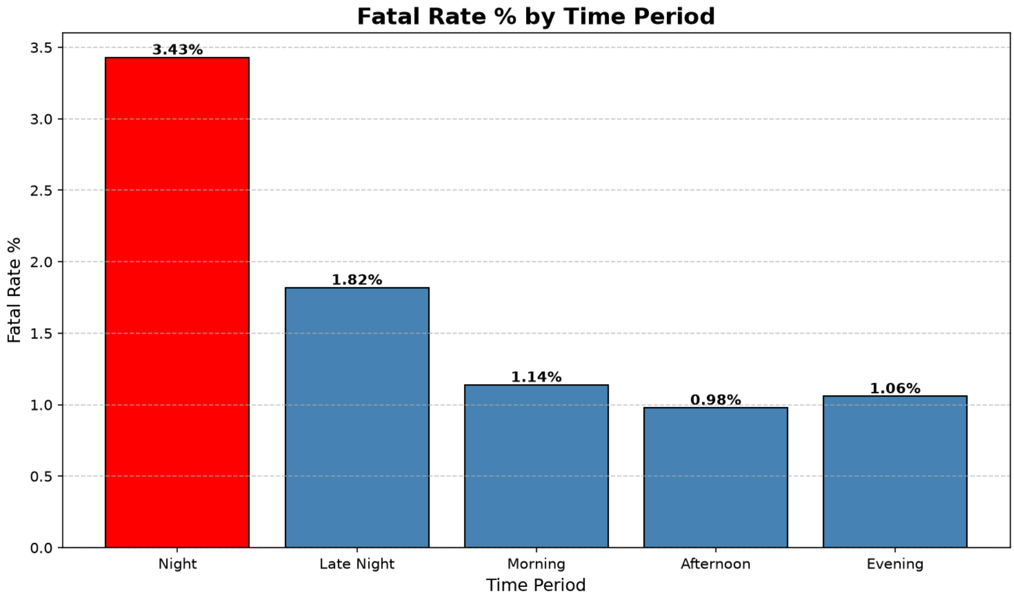

Figure 10: Fatal Rate % by Time Period — Night has highest fatal rate (3.43%)

Table 12: Top five hours by severity ratio

Hour	Severity ratio	Accidents
04:00	1.554	1,721
02:00	1.551	2,652
03:00	1.531	2,311
01:00	1.498	3,533
00:00	1.490	4,667

This finding is best illustrated by the 04:00 hour. With only 1,721 accidents it appears to be the safest hour of the day by frequency. Yet it carries the highest severity ratio of any hour at 1.554, meaning that when an accident does occur at 04:00 it is more likely to be serious or fatal than at any other time. This single example demonstrates clearly that frequency alone is not a sufficient measure of road safety risk, a quiet hour can be a deadly one.

Table 13: Weekday versus weekend frequency and severity

Week type	Accidents	Share	Severity ratio	Fatal rate	Fatalities
Weekday	232,848	75.6%	1.296	1.12%	2,614
Weekend	75,107	24.4%	1.377	1.78%	1,339

Although weekends account for only 24.4 per cent of all accidents, they produce 33.9 per cent of all fatalities. The weekend fatal rate of 1.78 per cent is approximately 59 per cent higher than the weekday rate of 1.12 per cent. This means that while fewer accidents happen on weekends, those that do are significantly more likely to result in death.

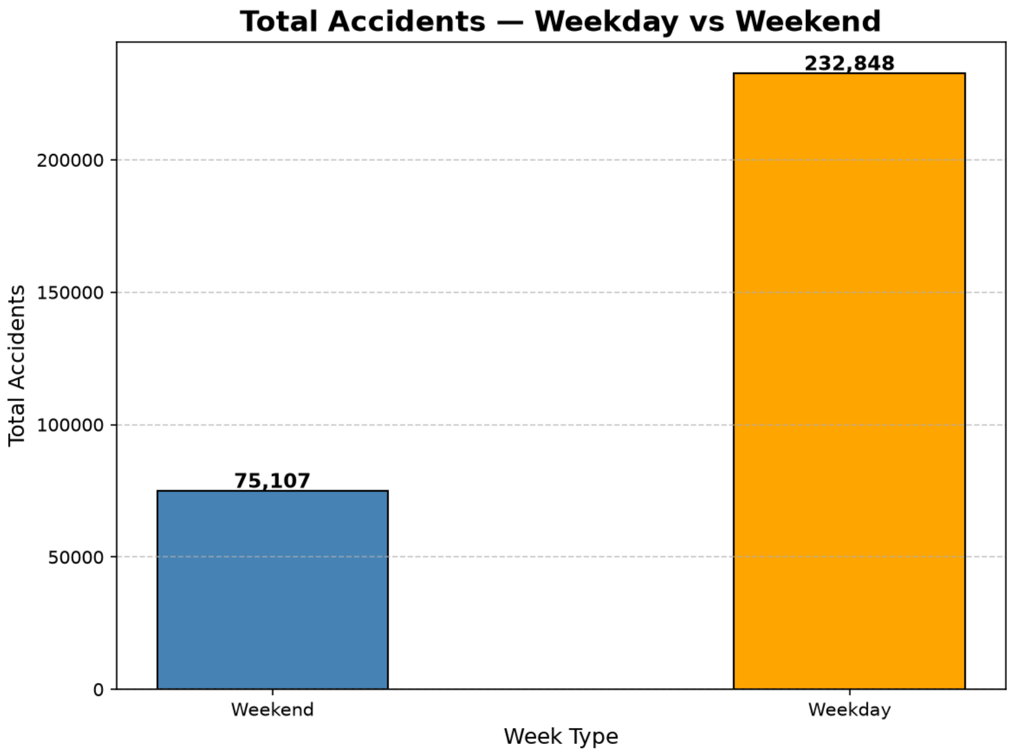

Figure 11: Total Accidents — Weekday vs Weekend

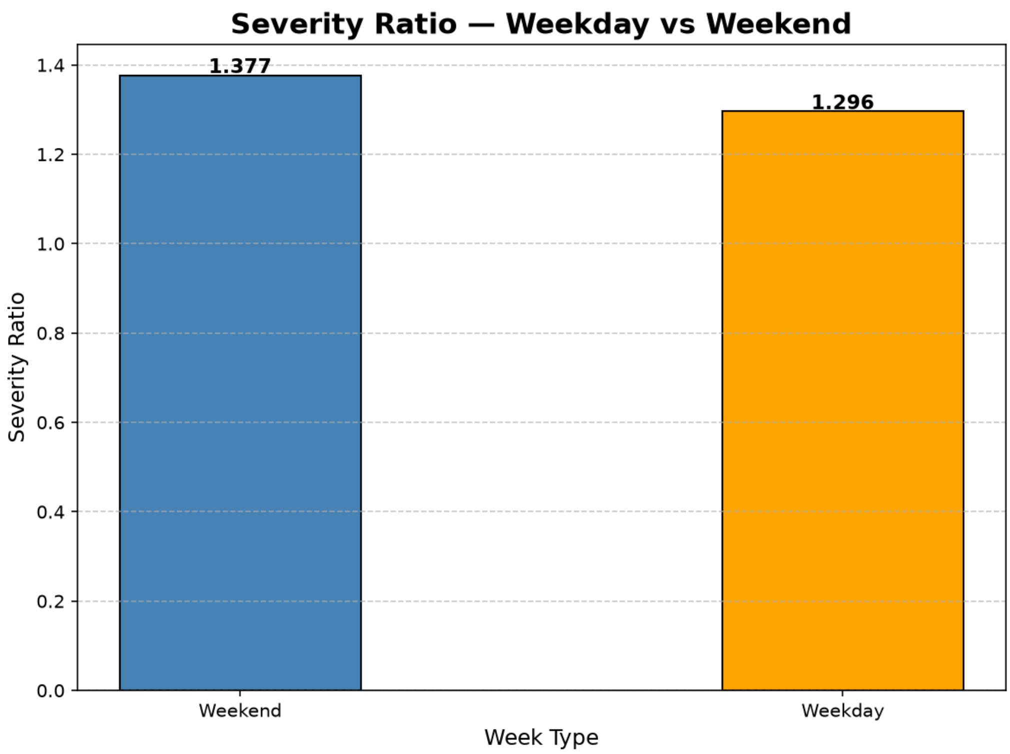

Figure 12: Severity Ratio — Weekend (1.377) higher than Weekday (1.296)

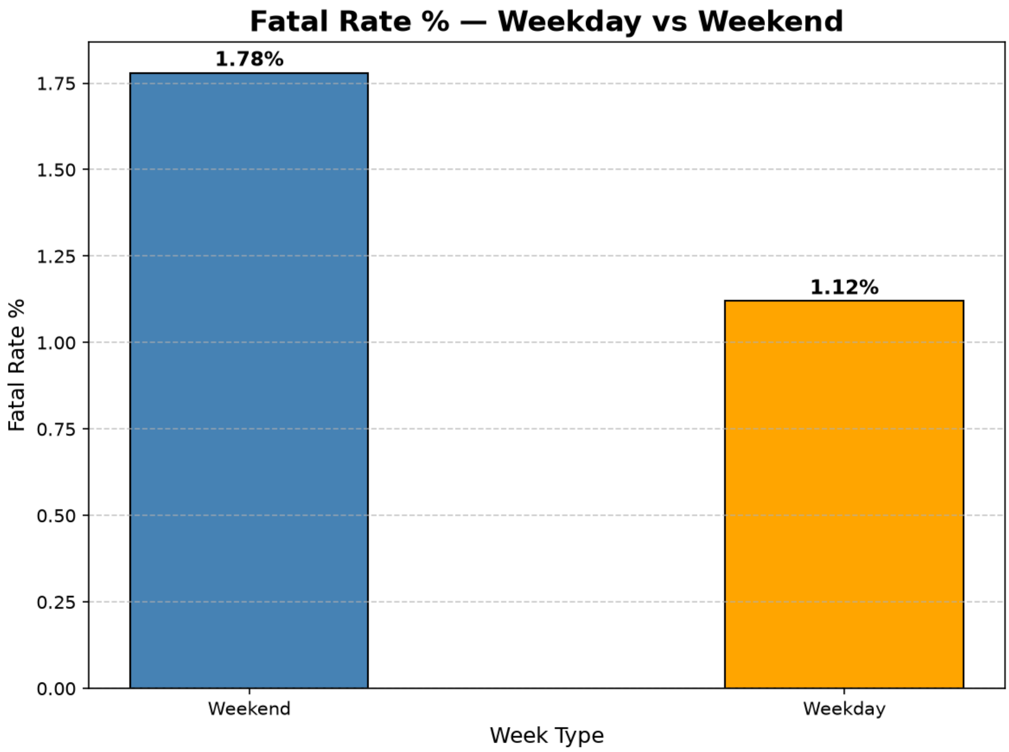

Figure 13: Fatal Rate % — Weekend (1.78%) higher than Weekday (1.12%)

These findings do not contradict the literature. Abdelwahab and Abdel-Aty (2001), Mujalli and de Oña (2011) and Delen et al. (2006) all find that time of day is a weak predictor of severity when other factors are controlled. RRSIS does not claim that the hour of day causes accidents to be more severe, it simply describes what the data shows: that accidents happening at night tend to have more serious outcomes than those happening in the evening. The cause is likely the conditions that come with night driving, reduced visibility, driver fatigue, higher speeds on empty roads and longer emergency response times, not the hour itself. For Rwanda the practical implication is clear regardless of the causal question. Night accounts for only 5.64 per cent of accidents but 15.1 per cent of all fatalities. A road safety strategy that ignores night hours because few accidents occur would be leaving the deadliest window unaddressed.

4.4 Severity Index Results

Table 14: Overall severity index distribution

Severity class	Weight	Accidents	Severity score	Share of severity
Slight	1	263,264	263,264	64.96%
Serious	3	40,738	122,214	30.16%
Fatal	5	3,953	19,765	4.88%
Total		**307,955**	**405,243**	**100.00%**

Applying the weights changes the picture significantly. Serious accidents rise from 13.23 per cent of total accidents to 30.16 per cent of the total severity score, and fatal accidents rise from 1.28 per cent to 4.88 per cent. This is exactly the intended effect, ensuring that rare but deadly accidents are not overwhelmed by the large number of slight accidents. The overall severity ratio for the entire dataset is 1.316, which serves as the baseline for comparing individual locations, road types and time periods throughout the analysis.

Table 15: Top five locations by severity score

Rank	Location	Accidents	Severity score	Severity ratio
1	Birmingham	6,165	7,805	1.266
2	Leeds	4,140	5,354	1.293
3	Westminster	2,811	4,341	1.544
4	Manchester	3,132	3,854	1.230
5	Bradford	3,006	3,840	1.277

Westminster provides the clearest example of why accident frequency and severity must be measured separately. It records 2,811 accidents, less than half of Birmingham's 6,165, yet it has 189 fatalities compared to Birmingham's 57, more than three times as many. As a result, Westminster's severity ratio of 1.544 far exceeds Birmingham's 1.266 and the dataset baseline of 1.316. Ranked by accident count, Birmingham is the leading problem and Westminster does not appear in the top two. Ranked by severity, Westminster is the most dangerous location in the dataset. The two rankings point to different interventions, Birmingham needs volume management while Westminster needs measures targeting the causes of fatal outcomes.

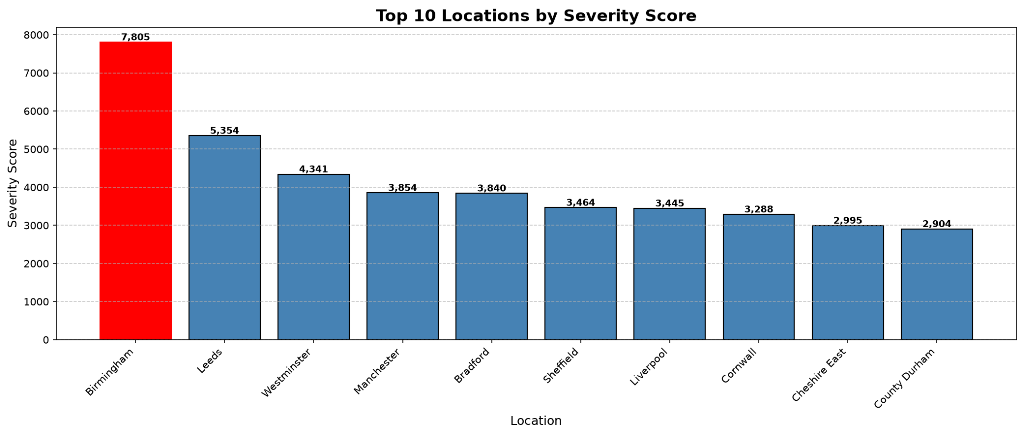

Figure 14: Top 10 Locations by Severity Score — Birmingham highest with 7,805

Table 16: Severity score by road type

Rank	Road type	Accidents	Severity score	Severity ratio
1	Single carriageway	230,595	307,409	1.333
2	Dual carriageway	45,466	59,414	1.307
3	Roundabout	20,929	24,801	1.185
4	One way street	6,197	7,889	1.273
5	Slip road	3,234	3,840	1.187

Single carriageway roads dominate the analysis, accounting for 74.9 per cent of all accidents and 77.4 per cent of the total severity score, approximately five times the severity score of dual carriageway roads. This is not surprising when the nature of the road type is considered: a single carriageway has no physical barrier separating traffic travelling in opposite directions. Any loss of control, misjudgment of speed or moment of distraction can result in a head-on collision, which is among the most severe accident types. This makes single carriageway roads a structural risk rather than a situational one, the danger is built into the road design itself.

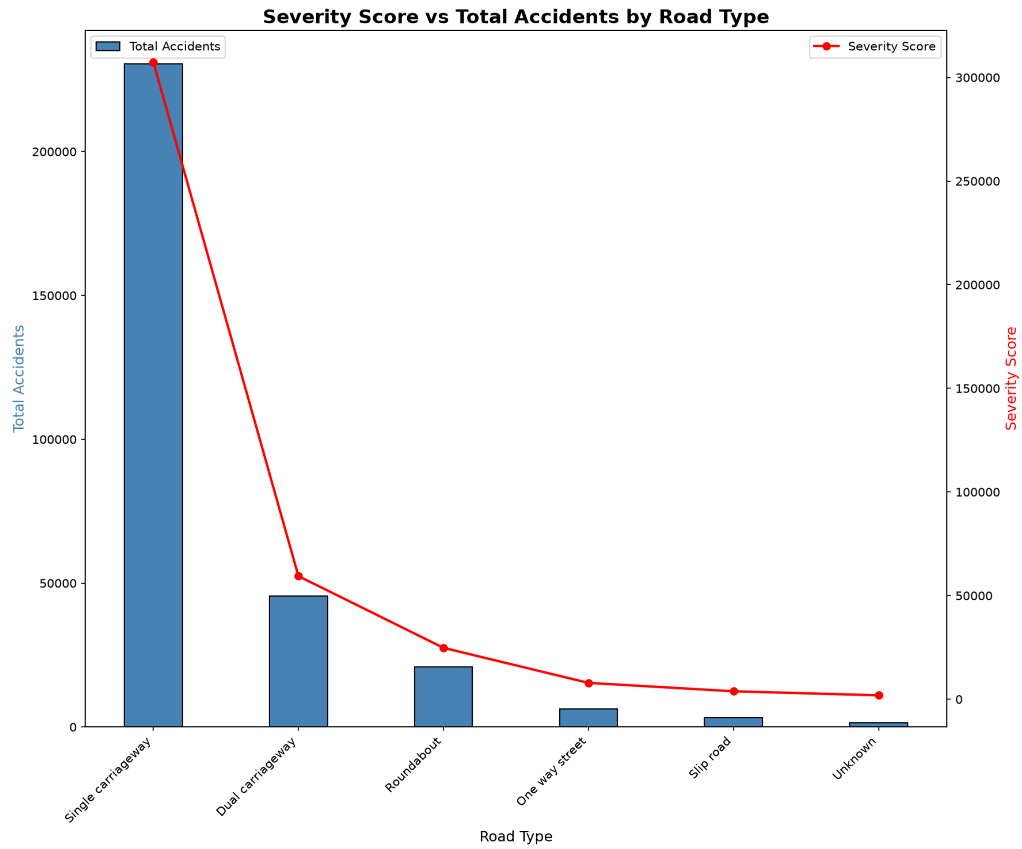

Figure 15: Severity Score vs Total Accidents by Road Type — Single carriageway dominates with severity 307,409

Table 17: Severity score by vehicle type

Rank	Vehicle type	Accidents	Severity score	Severity ratio
1	Car	239,780	315,470	1.316
2	Van / Goods 3.5t	15,692	20,774	1.324
3	Motorcycle 500cc+	11,226	14,790	1.317
4	Bus / Coach	8,686	11,300	1.301
5	Motorcycle 125cc	6,852	9,028	1.318

While vehicle type has a strong influence on how many accidents occur, cars alone account for 239,780 records, it has very little influence on how serious those accidents are. The severity ratios across all five vehicle classes fall within a narrow band of 1.301 to 1.324, meaning that once an accident happens, the type of vehicle involved makes almost no difference to the outcome. This aligns with Delen et al. (2006) [2], who also find vehicle type to be a weak severity predictor. One important qualification applies for Rwanda. The surrogate dataset is dominated by cars, which make up 77.9 per cent of records. Rwanda's vehicle fleet contains a far higher proportion of motorcycles, which are significantly more vulnerable in a collision due to the absence of any protective structure around the rider. The vehicle type finding from this analysis should therefore not be assumed to hold in the Rwanda context without verification against Rwanda-specific data.

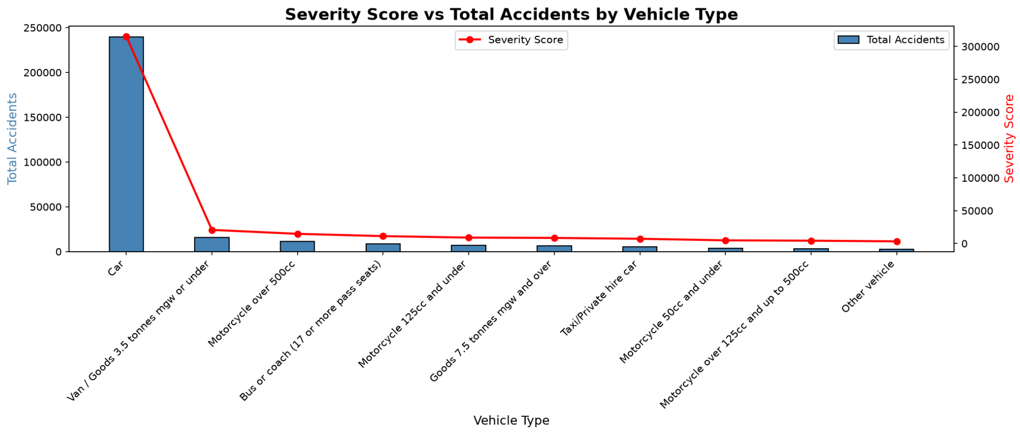

Figure 16: Severity Score vs Total Accidents by Vehicle Type — Cars account for 315,470 severity score

Table 18: Severity score by time period

Rank by severity score	Time period	Accidents	Severity score	Severity ratio
1	Evening	88,780	116,018	1.307
2	Morning	85,878	110,092	1.282
3	Afternoon	80,531	103,943	1.291
4	Late Night	35,391	48,939	1.383
5	Night	17,375	26,251	1.511

When time periods are ranked by total severity score, the ordering follows frequency, Evening ranks first and Night ranks last, simply because more accidents happen in the evening. However, when ranked by severity ratio, the order reverses completely, Night moves to first place and Evening to last. This single table makes visible the key insight from Savolainen et al. (2011): reducing the number of accidents and reducing the seriousness of accidents are two different problems that require two different strategies. An evening intervention targets frequency; a night intervention targets deadliness.

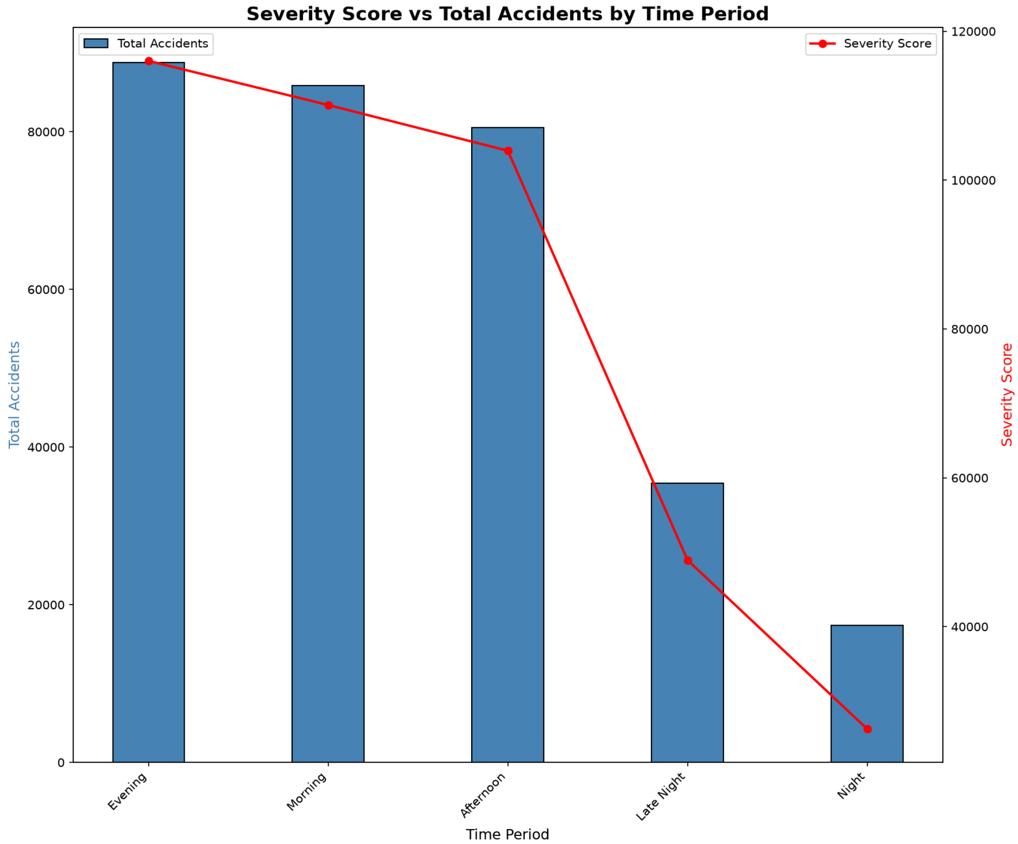

Figure 17: Severity Score vs Total Accidents by Time Period — Night highest severity ratio despite lowest frequency

4.5 Dangerous Factor Combination Results

Table 19: Top ten dangerous factor combinations by severity score

#	Road type	Speed	Weather	Surface	Period	Vehicle	Accidents	Severity
1	Single carr.	30 mph	Fine	Dry	Evening	Car	26,864	34,532
2	Single carr.	30 mph	Fine	Dry	Afternoon	Car	26,063	33,061
3	Single carr.	30 mph	Fine	Dry	Morning	Car	22,653	28,583
4	Single carr.	30 mph	Fine	Dry	Late Night	Car	9,394	12,776
5	Single carr.	60 mph	Fine	Dry	Evening	Car	4,935	7,947
6	Single carr.	60 mph	Fine	Dry	Afternoon	Car	4,866	7,760
7	Single carr.	60 mph	Fine	Dry	Morning	Car	4,590	6,930
8	Single carr.	30 mph	Fine	Dry	Night	Car	4,005	5,941
9	Single carr.	30 mph	Raining	Wet/damp	Evening	Car	4,582	5,756
10	Single carr.	30 mph	Fine	Wet/damp	Morning	Car	4,123	5,121

The most striking finding from the combination analysis is that eight of the top ten most dangerous combinations occur in fine weather with a dry road surface. This contradicts the common assumption that bad weather is the main driver of serious accidents, and it is consistent across all three severity papers reviewed. Abdelwahab and Abdel-Aty (2001) find weather statistically insignificant at $p = 0.556$; Mujalli and de Oña (2011) show that the probability of a serious outcome is actually higher in good weather (0.4653) than in heavy rain (0.3124); and Delen et al. (2006) excluded weather from all their models entirely. RRSIS now reproduces the same finding on 307,955 records using a completely different method. Speed limit is a far stronger differentiator of severity than weather. The 60 mph combinations return severity ratios of 1.610, 1.595 and 1.510, compared to 1.285, 1.268 and 1.262 for the equivalent 30 mph combinations. Per accident, the 60 mph combinations are approximately 20 to 26 per cent more harmful a direct consequence of the physics of collision, where injury severity increases with speed. Together the top ten combinations account for 36.4 per cent of all accidents in the dataset, meaning that a small and specific set of conditions is responsible for more than a third of the total accident burden. For Rwanda, the practical implication is clear: road safety intervention should not be concentrated on the rainy season. The most dangerous conditions are ordinary ones good weather, dry roads, a single carriageway, a car, during the evening. Enforcement and awareness campaigns should target everyday driving behavior, not adverse weather events.

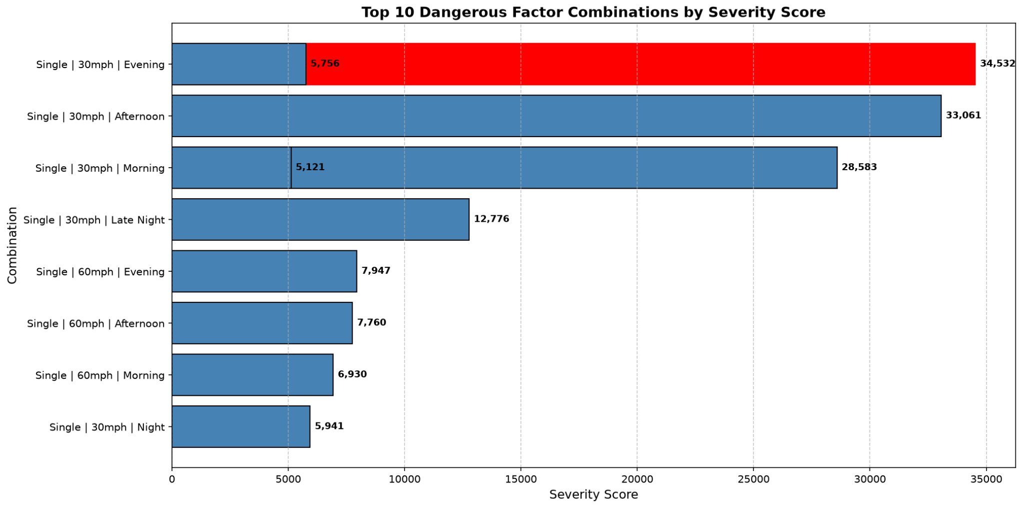

Figure 18: Top 10 Dangerous Factor Combinations by Severity Score — Single carriageway + 30mph + Fine weather + Evening + Car dominates

4.6 Window Function Location Ranking Results

Table 20: Top three locations per geographical category, ranked by dense_rank() on severity score

Category	Rank	Location	Accidents	Severity score	Severity ratio
Urban	1	Birmingham	6,099	7,717	1.265
Urban	2	Westminster	2,811	4,341	1.544
Urban	3	Leeds	3,329	4,231	1.271
Rural	1	Cornwall	1,992	2,584	1.297
Rural	2	County Durham	1,421	1,903	1.339
Rural	3	Wiltshire	1,158	1,816	1.568

Wiltshire provides an important example of why ranking within geographical categories matters. With only 1,158 accidents it would never appear in a national ranking dominated by large urban authorities. Yet its severity ratio of 1.568 is the second highest in the entire table, meaning that accidents in Wiltshire are more likely to be serious or fatal than in most other locations. It surfaces only because the window function forces rural locations to compete against other rural locations rather than against cities like Birmingham. This has a direct lesson for Rwanda. A national ranking based on accident counts alone would direct almost all resources to Kigali, where accident volumes are highest. But the road safety problem in Rwanda's provincial network — longer distances, higher speeds, limited lighting, longer emergency response times — is fundamentally different in character from the urban congestion problem in Kigali. Ranking urban and rural areas separately ensures that both problems receive attention and that genuinely dangerous rural roads are not made invisible by the scale of the capital city.

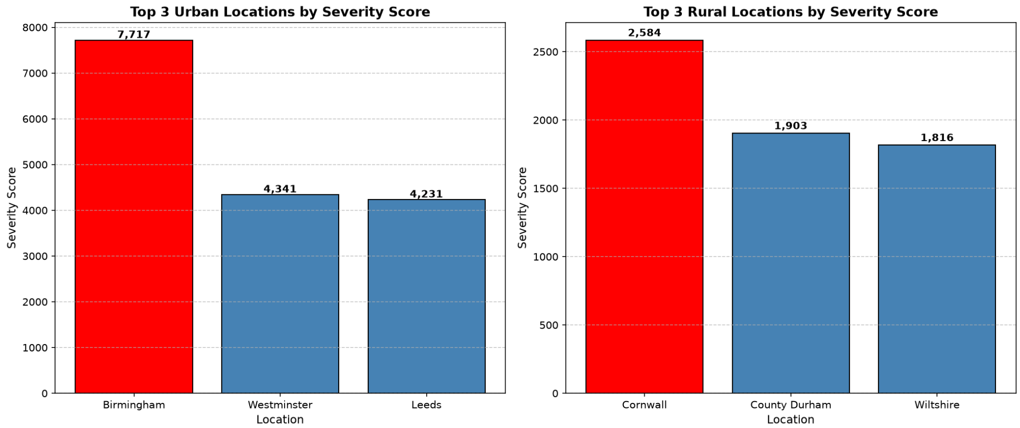

Figure 19: Top 3 Locations per Geographical Category (Urban/Rural) using dense_rank() Window Function

4.7 Risk Score Results

Table 21: Top ten locations by composite Road Safety Risk Score

Rank	Location	Accidents	Severity score	Fatal accidents	Risk score
1	Birmingham	6,165	7,805	57	0.7905
2	Westminster	2,811	4,341	189	0.6585
3	Leeds	4,140	5,354	38	0.5357
4	Bradford	3,006	3,840	30	0.3899
5	Manchester	3,132	3,854	24	0.3872
6	Camden	1,671	2,529	101	0.3701
7	Cornwall	2,606	3,288	43	0.3626
8	Liverpool	2,611	3,445	22	0.3376
9	Sheffield	2,750	3,464	16	0.3358
10	County Durham	2,228	2,904	28	0.3007

Westminster demonstrates exactly why a multi-dimensional score adds value over a simple accident count. It ranks only fifth by number of accidents, yet it rises to second place in the Risk Score at 0.6585 because it recorded 189 fatal accidents, the highest fatal count of any location in the dataset. Camden shows the same pattern, ranking sixth despite only 1,671 accidents because it has 101 fatalities. Sheffield illustrates the opposite: with 2,750 accidents it has more than Westminster and Camden, but only 16 fatalities push it down to ninth place. The fatal count component of the Risk Score is what separates these locations from one another. One limitation applies to all scores: they are calculated relative to the observed range in this dataset and do not include a traffic volume denominator. This means the scores measure the total accident burden at each location rather than the risk per vehicle passing through. A busy location will naturally accumulate more accidents than a quiet one even if both are equally dangerous per journey. This is an acknowledged limitation noted in Section 5.2.

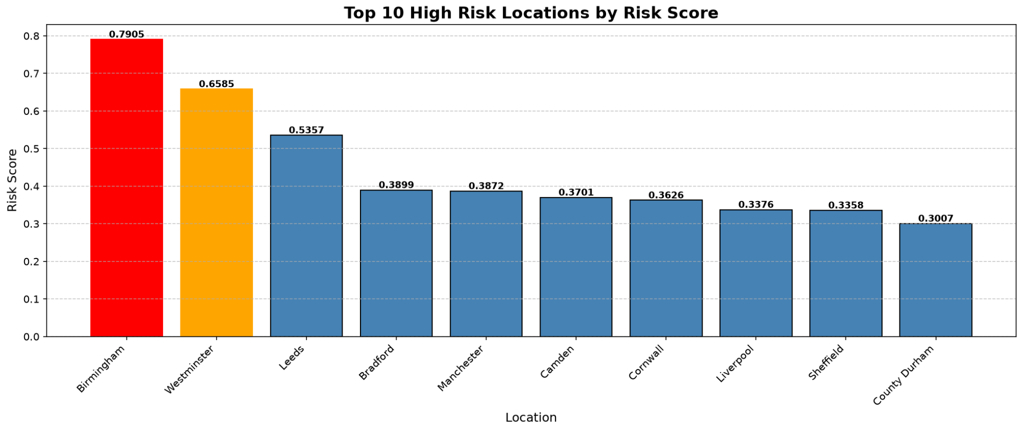

Figure 20: Top 10 High Risk Locations by Composite Risk Score — Birmingham (0.7905) and Westminster (0.6585) highest risk

4.8 Spark Performance Results

Table 22: Shuffle operations identified in the RRSIS pipeline

#	Triggering operation	Exchange type	Partitioning key	Partitions
1	groupBy	hashpartitioning	Local_Authority_(District)	200
2	orderBy	rangepartitioning	Severity_Score DESC	200
3	dropDuplicates	hashpartitioning	all columns	200

Every groupBy and orderBy operation in the pipeline requires Spark to redistribute records across the cluster so that related records end up on the same executor before being processed. This redistribution, known as a shuffle, is the most expensive operation in distributed computing because it involves moving data across the network. The most costly shuffle in this pipeline is `dropDuplicates()`, because Spark must compare every record against every other record using all columns as the comparison key, a much heavier operation than grouping by a single column like location or road type. To avoid repeating the costly ingestion and cleaning steps for every subsequent task, `cache()` was applied to `df_cleaned` after cleaning. This stores the cleaned dataset in memory so that Tasks 3 to 9 all read from memory rather than re-reading the raw CSV from HDFS and re-running all eight cleaning operations from scratch each time. The cache was confirmed successfully with 307,955 records stored, as shown in the screenshot `Task8_03_Cache_Usage.JPG` available on the project GitHub repository at https://github.com/Rubagumyabrave/Midterm_project_group_15.

4.9 Five Management Priorities

Priority 1 — Evening Rush Hour. *Evidence:* **72,929 accidents, 23.7 per cent of the dataset**, in the 16:00–18:00 window, with 17:00 alone at 26,964 accidents and severity 34,916. *Recommendation:* deploy traffic management during 16:00–18:00 on weekdays; in Rwanda, increase traffic police presence during evening hours in urban areas.

Priority 2 — Single Carriageway Roads. *Evidence:* **230,595 accidents (74.9 per cent) and severity 307,409**, five times the dual carriageway figure, present in all ten worst combinations. *Recommendation:* prioritise markings, speed cameras and lighting; in Rwanda, focus on rural roads connecting provincial cities.

Priority 3 — High Risk Locations. *Evidence:* **Birmingham 0.7905 with 57 fatal accidents, Westminster 0.6585 with 189, Leeds 0.5357 with 38**. *Recommendation:* concentrate resources in the top three rather than distributing evenly; in Rwanda, identify equivalent districts using RRSIS on national data.

Priority 4 — Friday and Weekend. *Evidence:* **Friday 50,529 accidents and 617 fatalities**, Saturday fewer accidents at 41,564 but 710 deaths, and a **weekend fatal rate of 1.78 against 1.12 per cent**. *Recommendation:* launch Friday and weekend campaigns; in Rwanda, target end-of-week fatigue through public awareness.

Priority 5 — Night Time. *Evidence:* **Night severity ratio 1.511 and fatal rate 3.43 per cent**, the highest of any period, with **04:00 at 1.554**, the highest single hour. *Recommendation:* improve lighting, night patrols and drink-driving enforcement; in Rwanda, critical for rural roads with limited lighting.

Table 23: Summary of the five management priorities

#	Priority	Principal evidence	Primary measure
1	Evening rush hour management	72,929 accidents (23.7%) in 16:00–18:00	Frequency
2	Single carriageway improvement	230,595 accidents, severity 307,409	Frequency and severity
3	High risk location intervention	Birmingham 0.7905, Westminster 0.6585	Composite risk score
4	Friday and weekend campaign	Friday 50,529; weekend fatal rate 1.78%	Frequency and fatal rate
5	Night time enhancement	Night ratio 1.511, fatal rate 3.43%	Severity ratio

5 Conclusion, Limitations and Recommendations

5.1 Conclusion

RRSIS processes 307,955 records on HDFS and PySpark, delivering four contributions: two-dimensional temporal analysis showing 04:00 lowest by count at 1,721 yet highest by severity ratio at 1.554; a risk score with literature-backed weights lifting Westminster to second place on 189 fatal accidents; combination analysis showing eight of the ten worst combinations in fine, dry conditions; and window ranking surfacing rural locations such as Wiltshire. It exceeds the literature at 307,955 records against a maximum of 30,358, retains 99.994 per cent of records against losses of 9.5 to 53 per cent, and delivers ranked interventions from a transparent auditable index rather than a black-box model. As a **descriptive prioritisation tool** built on a **surrogate dataset**, its findings transfer to Rwanda as mechanisms rather than magnitudes.

5.2 Limitations

- **L1 — Surrogate dataset.** The Kaggle dataset is not actual Rwanda accident data; all findings represent United Kingdom patterns from 2021–2022 and require validation against Rwanda-specific data.
- **L2 — No validation against held-out data.** Abdelwahab and Abdel-Aty (2001) report 60.4 per cent accuracy and Delen et al. (2006) 89.34 per cent, whereas RRSIS reports no accuracy or validation figure because it fits no model.
- **L3 — Fixed weights.** Delen et al. (2006) demonstrate predictor importance swinging from 0.5387 to 0.0043 across severity thresholds, whereas RRSIS uses fixed global weights that assume constant importance.
- **L4 — Temporal variables reframed.** Three independent studies find time-of-day and day-of-week weak severity predictors, so the RRSIS temporal analysis is presented as frequency and exposure analysis rather than severity explanation.
- **L5 — Spatial correlation.** Savolainen et al. (2011) note that adjacent locations share unobserved effects, so top-three rankings within the same corridor may reflect one underlying problem rather than three.
- **L6 — Underreporting bias.** Savolainen et al. (2011) report slight injuries undercounted 50–75 per cent, so locations dominated by slight crashes may be systematically under-ranked.

5.3 Future Recommendations

- Apply RRSIS to actual Rwanda National Police accident data, substituting only the HDFS input path and column mapping.
- Add a validation split, training on 2021 and testing against 2022, to establish whether the rankings are stable over time.
- Derive the severity and composite weights empirically through sensitivity analysis rather than assigning them by judgement.
- Add an exposure denominator such as traffic volume per location, converting the score from a burden measure to a rate measure.
- Extend the pipeline to real-time processing with Spark Structured Streaming to detect emerging accident concentrations.
- Integrate with Rwanda GIS mapping for corridor-level spatial analysis, addressing the spatial correlation limitation.

References

[1] Abdelwahab, H. T., & Abdel-Aty, M. A. (2001). Development of Artificial Neural Network Models to Predict Driver Injury Severity in Traffic Accidents at Signalized Intersections. *Transportation Research Record*, 1746, 6–13.

[2] Delen, D., Sharda, R., & Bessonov, M. (2006). Identifying significant predictors of injury severity in traffic accidents using a series of artificial neural networks. *Accident Analysis and Prevention*, 38(3), 434–444.

[3] Mujalli, R. O., & de Oña, J. (2011). A method for simplifying the analysis of traffic accidents injury severity on two-lane highways using Bayesian networks. *Journal of Safety Research*, 42(5), 317–326.

[4] National Institute of Statistics of Rwanda (NISR). (2024). *Statistical Yearbook 2024, Table 14.2.6 — Road Accidents*. Kigali: NISR.

[5] Rahm, E., & Do, H. H. (2000). Data Cleaning: Problems and Current Approaches. *IEEE Data Engineering Bulletin*, 23(4), 3–13.

[6] Rwanda National Police. (2023). *Road Accident Statistics 2023*. Kigali: Rwanda National Police.

[7] Savolainen, P. T., Mannering, F. L., Lord, D., & Quddus, M. A. (2011). The statistical analysis of highway crash-injury severities: A review and assessment of methodological alternatives. *Accident Analysis and Prevention*, 43(5), 1666–1676.

[8] Xavierberge. (2023). *Road Accident Dataset* [Data set]. Kaggle. https://www.kaggle.com/datasets/xavierberge/road-accident-dataset

[9] Rubagumya, B., Uwizeye, C., & Rwagasana, F. (2026). *RRSIS — Rwanda Road Safety Intelligence System: Midterm Project Group 15* [Source code, notebooks and evidence screenshots]. GitHub. https://github.com/Rubagumyabrave/Midterm_project_group_15

Appendix A — Complete Screenshot List

All evidence screenshots are available on the project GitHub repository at:
https://github.com/Rubagumyabrave/Midterm_project_group_15
They are not embedded in the report body.

#	Screenshot Filename	Task
1	Pyspark_1.JPG	Task 1
2	Pyspark_2.JPG	Task 1
3	Task1_01_HDFS_mkdir.JPG	Task 1
4	Task1_02_HDFS_put_upload.JPG	Task 1
5	Task1_03_HDFS_put_upload.JPG	Task 1
6	Task1_06_Dataset_Schema.JPG	Task 1
7	Task1_07_Record_Column_Count.JPG	Task 1
8	Task1_08_Sample_Records.JPG	Task 1
9	Task2_01_Missing_Values.JPG	Task 2
10	Task2_02_Duplicate_Records.JPG	Task 2
11	Task2_03_Data_Types_Check.JPG	Task 2
12	Task2_04_Suspicious_Values.JPG	Task 2
13	Task2_05_After_Cleaning_Count.JPG	Task 2
14	Task3_01_Accidents_By_Hour.JPG	Task 3
15	Task3_02_Accidents_By_Day.JPG	Task 3
16	Task3_03_Accidents_By_Month.JPG	Task 3
17	Task3_04_Weekday_Vs_Weekend.JPG	Task 3
18	Task3_05_Accidents_By_Time_Period.JPG	Task 3
19	Task3_05_Top5_HighRisk_Periods.JPG	Task 3
20	Task3_06_Hour_Severity_Ratio.JPG	Task 3
21	Task3_07_Time_Period_Severity_Ratio.JPG	Task 3
22	Task3_10_Weekday_Weekend_Severity.JPG	Task 3
23	Task4_00_Top_Severity_classification.JPG	Task 4
24	Task4_01_overarall_severity_score.JPG	Task 4
25	Task4_02_Top_Locations_By_Severity.JPG	Task 4
26	Task4_03_Severity_By_Road_Type.JPG	Task 4
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28	Task4_05_Severity_By_Time_Period.JPG	Task 4
29	Task5_01_Top10_Dangerous_Combinations.JPG	Task 5
30	Task7_03_Final_Ranked_Locations.JPG	Task 7
31	Task8_03_Cache_Usage.JPG	Task 8